MIDAS: An Energy-Efficient Microscaling Digital Compute-In-Memory-Based Accelerator with Spatio-Temporal Cyclic Alignment for Generative AI Inference

Jiwon Choi, Seongyon Hong, Wooyoung Jo, Wonhoon Park, Sunjoo Whang, Sangjin Kim, and Hoi-Jun Yoo

KAIST, Daejeon, Korea

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Generative AI (GenAI), including large language models (LLMs) and diffusion models, is rapidly advancing with substantial deployment demand. However, their large memory footprints and high computational costs constrain deployment on edge devices. Digital compute-in-memory (DCIM) accelerators have emerged as a promising solution for energy-efficient GenAI processing [1-4]. Meanwhile, various low bit-width data formats have been proposed for GenAI. However, outliers in weights and activations pose significant quantization challenges. Microscaling (MX) format, a standardized sub-8-bit representation, constrains the quantization range within each block, thereby mitigating the impact of outliers while maintaining accuracy with reduced bit-width [5,6].

However, existing DCIMs cannot fully exploit the MX format due to three key challenges. First, cell under-utilization from mantissa alignment becomes severe with the MX format, since mantissa truncation, a commonly used method to mitigate this issue, causes significant accuracy degradation in the sub-8-bit regime. Second, exponent handling, such as maximum exponent searching and activation mantissa alignment, shows an increased portion of energy and area as the bit-width decreases. The portion of the energy-area product attributed to exponent handling logic in the MXFP4 is 45.4%p higher than FP16. Third, the hybrid computing scheme used in existing DCIMs [1,7,8] conflicts with the block-wise shared scale (SS) of MX, causing severe accuracy degradation. To address these challenges, we propose MIDAS, an MX-optimized DCIM-based GenAI accelerator delivering both high energy and area efficiency. First, a spatio-temporal cyclic alignment (STCA) and sign-magnitude (SM)-based assembled column addition (SACA) enable compact weight mapping without redundant computation, improving energy and area efficiency by 1.72× and 2.47×, respectively. Second, MX-aware minimum exponent post-detection (MEPD) reduces power and area overhead in exponent handling by 68.8% and 63.2%, respectively. Third, a sense amplifier-based dynamic bit-significance handling (SADB) with shared scale-aware channel regrouping (SSCR) further improves energy efficiency by 27.3%.

Fig. 2 shows the overall architecture of MIDAS. It comprises two 256 KB global memories (GMEMs), two special function engines (SFEs), a 2D mesh network-on-chip, and four MX CIM engines (MXCEs). Each MXCE contains a shared-scale-aware channel regrouping unit (SCRU), a 24 KB input memory, an index-aware accumulation unit (IAU), and six 16 kb cyclic alignment-based CIM macros (CA-CIMs) for weight-stationary computation. A CA-CIM includes a cyclic input buffer, a global counter, a relative counter generator (RCG), a minimum exponent post-detection unit (MEPU), and 16 local cell array (LCA) columns. Each LCA column accumulates 32 weight channels and consists of an MSB partial product adder tree (MPAT), an LSB partial product adder tree (LPAT), and 32 LCAs. Each LCA has a 32×1 8T SRAM cell array and peripheral logic. Bitwise multiplication results are simultaneously read from the bit lines, summed by peripheral logic, and forwarded to the MPAT and LPAT.

Fig. 3 illustrates the detailed STCA with SACA dataflow. In STCA, both activations and weights are mapped in a column-major manner and are cyclically shifted by their respective exponent values when written into the cyclic input buffer (CIB) and each LCA. During execution, activations are cyclically shifted by one bit per cycle within the CIB before being fed into the LCAs. By independently accounting for activation and weight exponents, STCA enables each activation to be broadcast across weights of different output channels, thereby maximizing activation reusability. Furthermore, due to the cyclic pre-alignment that ensures compact storage, STCA achieves high area efficiency. SACA is further devised to improve energy efficiency. Since the sign of each partial product (PP) column in an LCA is identical, we adopt SM representation for computation, eliminating redundant operations caused by sign extension in conventional two's complement. SACA also reduces latency by shifting the least-significant-bit (LSB) PP columns and assembling them with the most-significant-bit (MSB) PP columns. To support additions of PPs with different scales within each PP column, each LCA employs a multiplexed full adder (MFA). Since STCA continuously streams activations regardless of whether the PP columns being produced in a given cycle are valid, every LCA integrates a valid region detection unit (VRDU) to identify valid PP columns and decides whether the results should be forwarded to the LPAT and MPAT. Together, STCA and SACA achieve 1.72× higher energy efficiency and 2.47× better area efficiency compared to the conventional FP DCIM baseline [1].

Fig. 4 shows the proposed novel exponent processing pipeline with two-stage optimization. First, the temporal alignment offset reformulation (TAOR) introduces a relative counter to eliminate redundant exponent computations. Unlike the conventional approach, which computes offsets for every activation-weight pair, TAOR incorporates the activation exponent into a relative counter, exploiting activation broadcast along the output channel direction. This reduces the required logic per LCA to a single 4-bit subtractor and a single D flip-flop. Second, the MEPD utilizes AND-trees to reduce energy and area overhead in exponent range detection. MEPD detects the minimum exponent rather than the maximum since near-minimum exponents are more sparsely distributed under MX quantization. The end signals from each VRDU are aggregated through an AND-tree, with the final output indicating the minimum exponent point and triggering early stopping. Furthermore, NOR operations on row-wise valid signals detect invalid PP columns, enabling word-line gating. Hence, the proposed novel exponent processing pipeline reduces power, area, and latency by 68.8%, 63.2%, and 82.2%, respectively.

Fig. 5 shows the two operation modes of SADB and the detailed SSCR scheme in MXCE. SADB enables flexible computation modes within an LCA by adjusting sense amplifier reference voltages. In accurate mode, three cells per LCA are accessed through separate bit-lines, performing full-adder operations. In approximate mode, two cells are accessed through the same bit-line, and the sense amplifier performs bit-wise OR or AND operations using higher or lower reference voltages, respectively. This adapts to both block-wise shared scale and intra-block bit position, enabling fine-grained bit-significance handling for high energy efficiency with minimal accuracy loss. SSCR improves energy efficiency by regrouping blocks with similar SS along the output channel dimension, increasing the proportion of approximate computations. Blocks are categorized into high- or low-SS groups via SCRU, allocated separately within the CA-CIM, and post-computation results are reordered by the IAU. SADB and SSCR schemes achieve a 27.3% improvement in energy efficiency with negligible SQNR loss.

Fig. 6 shows the benchmark results for GenAI tasks, especially with LLMs, and a comparison with previous DCIMs. MIDAS achieves system energy efficiencies of 28.2 TFLOPS/W, 16.2 TFLOPS/W, and 17.3 TFLOPS/W on GPT2-Large, Llama3.2-1B, and Qwen2-1.5B, respectively. With the MXFP6 format on GPT2-Large, MIDAS improves system energy efficiency by 2.71× over the baseline. Moreover, MIDAS achieves 1.58× higher system energy efficiency and 3.32× higher macro energy efficiency over the state-of-the-art DCIM [2]. Unlike previous MX-supported DCIM [4], which focused on the MXINT format, this work targets MXFP optimization. MIDAS is fabricated in 28 nm CMOS technology with a 7.56 mm^2 area, as shown in Fig. 7. It operates at 0.85-1.0 V supply voltage under 50 MHz-250 MHz clock frequency. Its measured macro energy and area efficiency is 96.3 TFLOPS/W and 0.69 TFLOPS/mm^2 at MXFP4, with system energy efficiency of 28.2 TFLOPS/W and 17.3 TFLOPS/W for MXFP4 and MXFP6, respectively. In conclusion, MIDAS enables energy and area-efficient GenAI processing by optimizing DCIM architecture for sub-8-bit MX format.

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References:

[1] A. Guo et al. "A 28nm 64-kb 31.6-TFLOPS/W Digital-Domain Floating-Point-Computing-Unit and Double-Bit 6T-SRAM Computing-in-Memory Macro for Floating-Point CNNs," ISSCC 2023: 128-130

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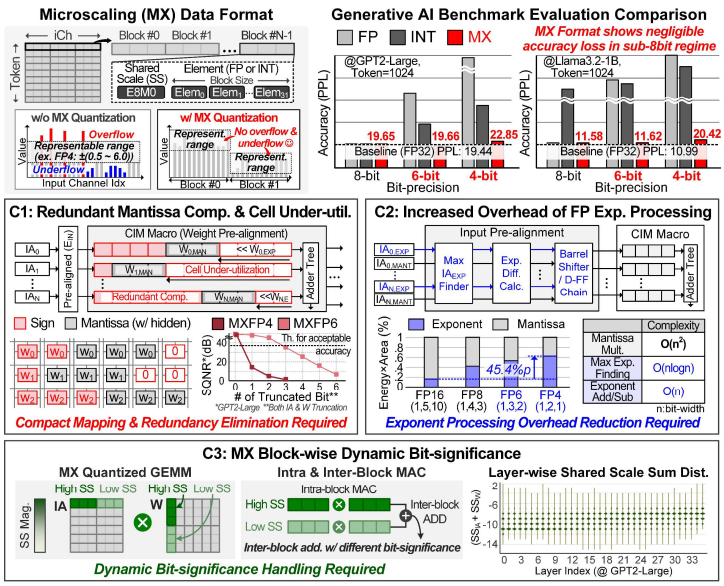


Fig. 1. Challenges of Microscaling CIM accelerator.

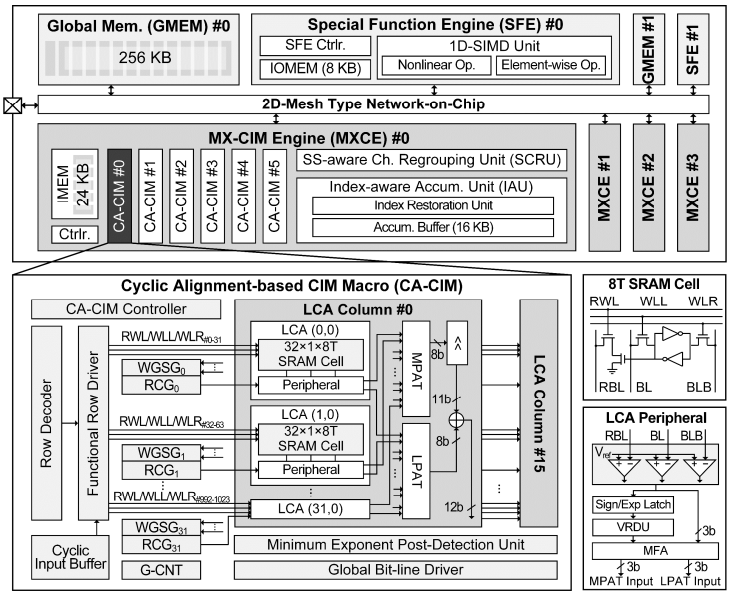


Fig. 2. Overall architecture of MIDAS.

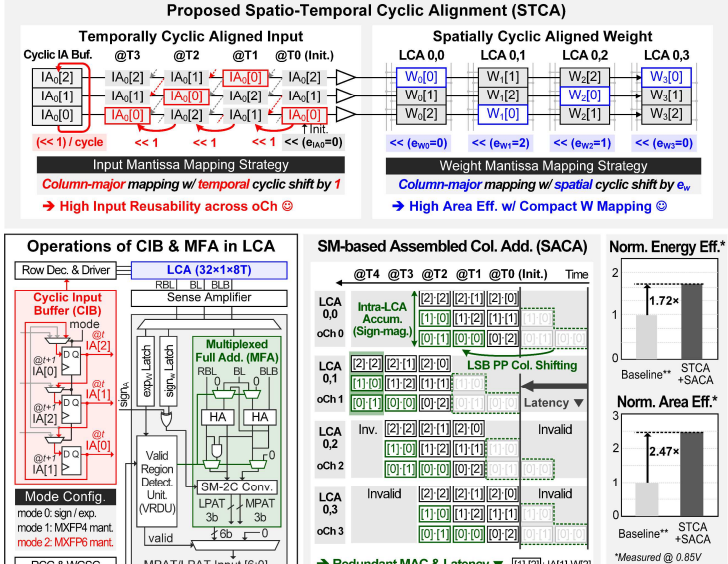


Fig. 3. Proposed STCA and SACA dataflow.

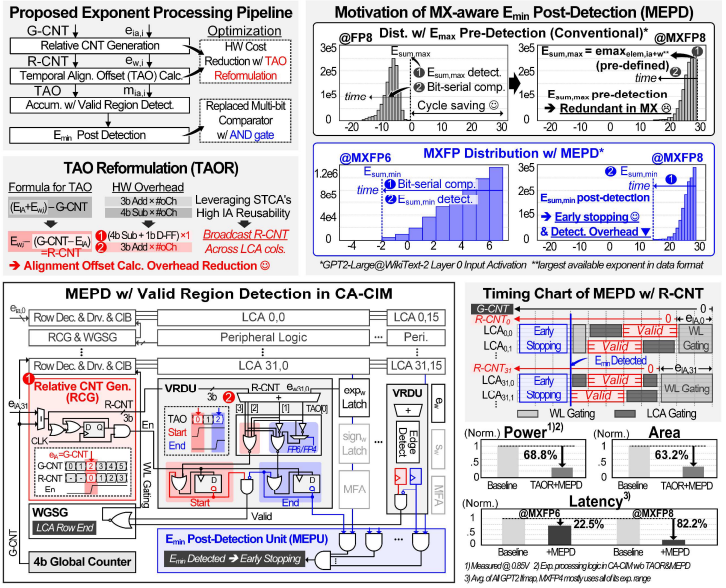


Fig. 4. Operations of TAOR and MEPD in CA-CIM.

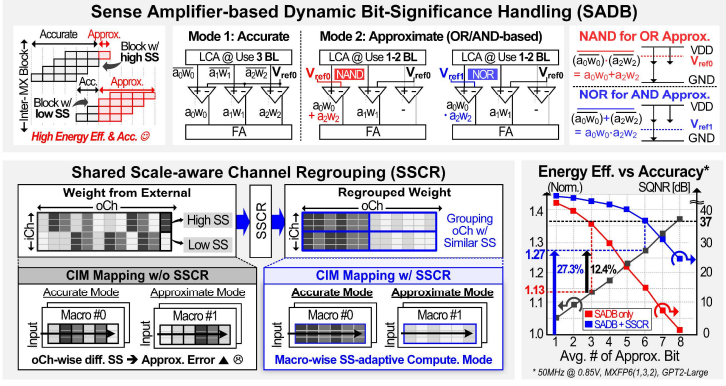
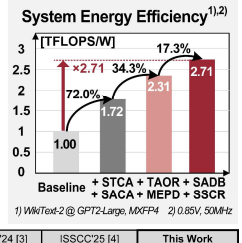


Fig. 5. Operation of SADB with SSCR scheme.

Benchmark Results for Generative AI Tasks

	Large Language Model		
Model	GPT2-Large	Llama3.2-1B	Qwen2-1.5B
Dataset	WikiText-2	WikiText-2	WikiText-2
Parameter	774M	1B	1.5B
Precision[1](IA/W)	MXFP4 / MXFP4[2]	MXFP6 / MXFP6	MXFP6 / MXFP6
Accuracy (PPL)	20.11 (GT:19.44)	11.62 (GT: 10.99)	11.12 (GT: 10.70)
System Energy-Efficiency[2] (TFLOPS/W)	28.2	16.2	17.3

1) MXFP8 for attention block operations(QK^T, AV) 2) MXFP6 for projection layer in attention block 3) 0.85V, 50MHz



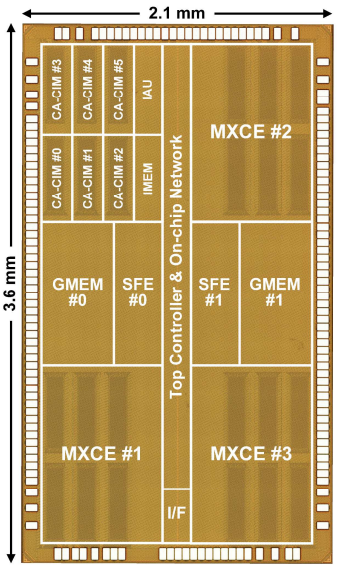
1) WikiText-2 @ GPT2-Large, MXFP4 2) 0.85V, 50MHz

	ISSCC'23 [7]	ISSCC'23 [1]	ISSCC'24 [2]	ISSCC'24 [3]	ISSCC'25 [4]	This Work
Technology	22 nm	28 nm	28 nm	28 nm	16 nm	**28 nm**
MX Support	X	X	X	X	O	**O**
Target Model	CNN	CNN	CNN/Transformer	CNN/Transformer	CNN	**Transformer**
Target Task	Image Classification	Image Classification/ Detection/Segmentation	Image Classification/ Text Generation	Image Generation	Image Classification	**Generative AI**
Data Precision	BF16	BF16/INT8	POSIT16/ POSIT8	INT10/INT16/ FP16/BF16	INT8/BF16/ MXINT8	**MXFP4/MXFP6/ MXFP8**
Die Area (mm^2)	18	0.146	1.41	3.67	8	**7.56**
Macro Area (mm^2)	18	0.146	NA	0.225 / 0.027	8	**0.062**
Supply Voltage (V)	0.6-0.8	0.6-0.9	0.55-1.0	0.6-1.0	0.4-0.8	**0.85 – 1.0**
Frequency (MHz)	NA	NA	0.55-1.0	50-540	NA	**50 – 250**
Macro Area Efficiency (TFLOPS/mm^2)	0.069-0.071	1.13-2.05	NA	NA	5.8[5]@BF16 11.4[5]@MXINT8	**0.69 @MXFP4 0.52 @MXFP6**
Macro Energy Efficiency (TFLOPS/W)	16.2-70.2[1]	14-31.6@BFP16[2] 19.5-44@INT8[2]	16.3-83.2@POSIT8[2,3] 7.5-38.4@POSIT16[2,3]	80.3@FP16[4]	91.9[1,5]@BF16 133.5[1,5]@MXINT8	**96.3 @MXFP4 54.1 @MXFP6**
System Energy Efficiency (TFLOPS/W)	-	-	10.9-55.6@POSIT8[2,3] 5.4-27.6@POSIT16[2,3]	74.3@BF16[4] 67.9@FP16[4]	-	**28.2 @MXFP4 17.3 @MXFP6**

1) Evaluated w/ 90% input sparsity 2) Evaluated w/ 50% input sparsity 3) 0.55V, 78MHz 4) 0.65V, 120MHz 5) 0.4V, 10% input toggle rate
6) 0.8V 7) Claiming of supporting MXFP w/o providing detialled performance 8) Claiming of supporting MXFP w/o providing detialled performance

Fig. 6. Measurement result and comparison table.

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	Specifications	
Technology	Samsung 28 nm	
Chip Area	7.56 mm^2	
Macro Area	0.062 mm^2	
Supply Voltage (V)	0.85-1.0	
Clock Frequency	50-250 MHz	
Data Precision	MXFP4 (1,2,1)	MXFP6 (1,3,2)
Macro Performance[1,2]		
Macro Area Efficiency[3] [TFLOPS/mm^2]	0.69	0.52
Macro Energy Efficiency[4] [TFLOPS/W]	96.3	54.1
System Performance[1,2]		
System Energy Efficiency[4] [TFLOPS/W]	28.2	17.3

1) 1 MAC = 2 Ops *2) WikiText-2 @ GPT2-Large*
3) 1.0V, 250MHz *4) 0.85V, 50MHz*

Fig. 7. Chip photograph and performance summary.

Additional References:

[2] Y. Wang, et al. ”34.1 A 28nm 83.23 TFLOPS/W POSIT-Based Compute in-Memory Macro for High-Accuracy AI Applications,” ISSCC 2024: 566-568.

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[4] WS. Khwa et al. ”14.2 A 16nm 216kb, 188.4 TOPS/W and 133.5 TFLOPS/W Microscaling Multi-Mode Gain-Cell CIM Macro Edge AI Devices,” ISSCC 2025: 1-3.

[5] BD Rouhani, et al. "Microscaling data formats for deep learning," arXiv preprint arXiv:2310.10537, 2023.

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[8] A. Guo et al. ”34.3 A 22nm 64kb Lightning-Like Hybrid Computing in-Memory Macro with a Compressed Adder Tree and Analog-Storage Quantizers for Transformer and CNNs,” ISSCC 2024: 570-572

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